

Scrum Fundamentals

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Form into Cross-Functional Teams



- Write your first name on a name tag
- Add one or more stars to your name tag:
 - ★ You have experience coding/programming
 - ★ You have experience using/testing software
 - ★ You have experience designing/architecting software
 - ★ You have experience managing a software project
 - ★ You have Scrum experience (or have read the Scrum guide)
- Form into a cross-functional, co-located team
- Introduce yourself to your team members



10 Minutes

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Richard Hundhausen

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Agenda

1. Scrum is popular
2. Scrumdamentals
3. Using Scrum
 - > Break
4. Continuous Improvement
5. Next Steps

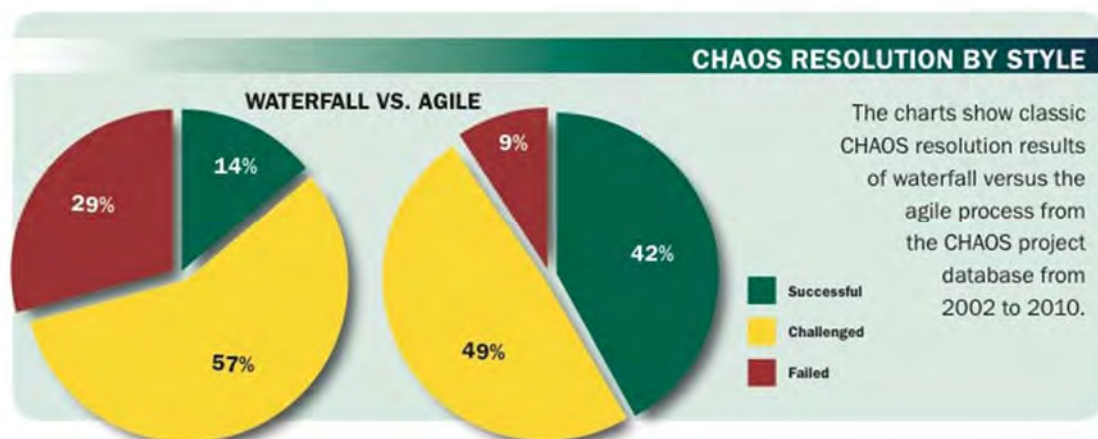
(Ask questions at any time)

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Scrum is popular (because it works)

The CHAOS Manifesto (2011)



The CHAOS Manifesto, Copyright 2011

The FBI Automated Case Support (ACS) program

- A CRM for criminals
- Built using 1970s era software tools (Natural, ADABAS, ...)
- Most believed it was obsolete when it shipped in 1995

The FBI Automated Case Support (ACS) program

- First, it was to be replaced by the *Virtual Case File*
- Estimate: 3 years at \$380 million
- Coding started in 2001
- Coding scrapped in 2005
- Wasted: \$170 million

The FBI Automated Case Support (ACS) program

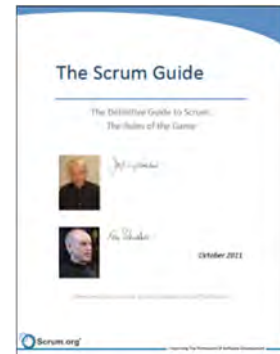
- Next, it was to be replaced by *Sentinel*
- Estimate: 4 phases over 6 years at \$450 million
- Coding started in 2006
- Contract canceled in 2010 after only 2 phases
- Wasted: \$405 million

The FBI Automated Case Support (ACS) program

- Project reset and a Scrum studio was setup in the basement of the Hoover building
- Staff reduced from 400 to 40
- 1 year and \$30 million:
- Code complete

Scrum is ...

- An framework for Agile software development
- A set of rules defined in the Scrum Guide
- Easy to learn but difficult to master



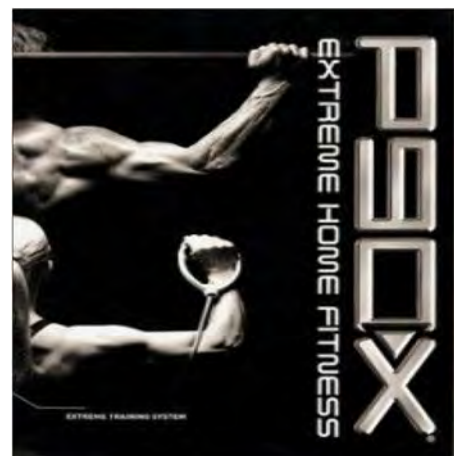
<http://www.scrum.org>

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Scrum is to Agile ...

- As P90X® is to "getting healthy"
- Both have core principles:
 - Eat better and exercise
- P90X® is a framework:
 - Daily menus and specific exercises
- In both cases, you are still the one that has to do it every day!



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Scrumdamentals

Roles, Events, Artifacts

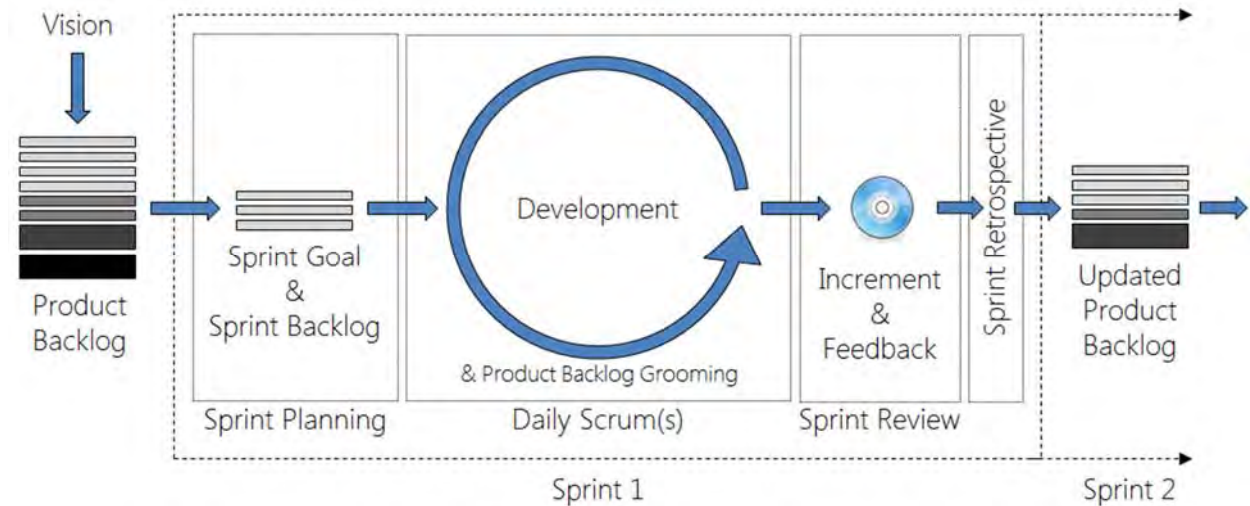
Roles
Product Owner
Scrum Master
Development Team

Events
The Sprint
Sprint Planning Meeting
Daily Scrum
Sprint Review
Sprint Retrospective

Note: All Events are timeboxed

Artifacts
Product Backlog
Sprint Backlog
The Increment (of Working Software)

The Scrum Process



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Role: Product Owner

- The Product Owner is responsible for maximizing the value of the product and the work of the Development Team

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Role: Scrum Master

- The Scrum Master is responsible for ensuring Scrum is understood and enacted

Role: Development Team

- The Development Team consists of professionals who do the work of delivering a potentially releasable Increment of "Done" product at the end of each Sprint

Event: The Sprint

- The heart of Scrum is a Sprint, a time-box of one month or less during which a “Done”, useable, and potentially releasable product Increment is created

Event: Sprint Planning Meeting

- The work to be performed in the Sprint is planned at the Sprint Planning Meeting and the plan is created by the collaborative work of the entire Scrum Team

Event: Daily Scrum

- The Daily Scrum is a 15-minute time-boxed event for the Development Team to synchronize activities and create a plan for the next 24 hours

Event: Sprint Review

- A Sprint Review is held at the end of the Sprint to inspect the Increment and adapt the Product Backlog if needed

Event: Sprint Retrospective

- The Sprint Retrospective is an opportunity for the Scrum Team to inspect itself and create a plan for improvements to be enacted during the next Sprint

Artifact: Product Backlog

- The Product Backlog is an ordered list of everything that might be needed in the product and is the single source of requirements for any changes to be made to the product

Artifact: Sprint Backlog

- The Sprint Backlog is the set of Product Backlog items selected for the Sprint plus a plan for delivering the product Increment and realizing the Sprint Goal

Artifact: The Increment

- The Increment is the sum of all the Product Backlog items completed (according to the definition of "Done") during a Sprint and all previous Sprints

Using Scrum

First, You Need a Product

- Think in terms of a *product*, not a project
- What are the product's success criteria?

The Product Backlog ...

- Contains a list of product “desirements”
 - Features, enhancements, behaviors, bugs
- Product Backlog items ...
 - Are clearly expressed (the “what”, not the “how”)
 - Have a business value
 - Are ordered (prioritized) to best achieve goals
 - Should be visible, transparent, and clear to all

The User Story format works well ...

- A way to write Product Backlog items:

As a (role) I want (something) so that (benefit)

- Example:

As a website visitor, I want to see a list of recent tweets, so that I know that the company and its products are alive and well

Example Product Backlog

The diagram shows a Product Backlog table with the following columns: ID, Title, Description, Area, Value, Effort, and Priority. Callouts explain the types of items and who provides them:

- Feature:** Indicated by a green box pointing to the 'Twitter feed' item (ID 3).
- Bug:** Indicated by a green box pointing to the 'Wrong sales tax' item (ID 4).
- Epic:** Indicated by a green box pointing to the 'Mobile-friendly site' item (ID 5).
- Anyone can provide this:** An orange callout pointing to the 'Customer Login' item (ID 1).
- Product Owner provides this:** Two orange callouts pointing to the 'Product return' (ID 2) and 'Twitter feed' (ID 3) items.
- Development Team provides this estimate:** An orange callout pointing to the 'Value' and 'Effort' columns.

ID	Title	Description	Area	Value	Effort	Priority
1	Customer Login	As a returning ...	Admin	45	8	2
2	Product return	As an unhappy ...	Products	75	13	4
3	Twitter feed	As a visitor to ...	Marketing	25	5	3
4	Wrong sales tax	As a purchaser ...	Orders	90	3	1
5	Mobile-friendly site	As a mobile user ...	*	60	34	
...						

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Acceptance Criteria

- In Scrum, the Acceptance Criteria are the requirements
 - They enumerate what the Product Owner expects and what the Development Team needs to accomplish
- Examples:
 - UX should target 1024 x 768 monitors
 - Internet Explorer 8.0+ should be supported
 - Instructions should be in English and Spanish

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Business Value

- Typically an integer value
 - Items with a larger integer have more business value than an item with a smaller integer value
 - The Product Owner determines the business value and the scale
- Examples:
 - Range of 1-100
 - Range of 1-n where the numbers represent strategic alignment, reduction in costs, competitive advantage, generates revenue, generates awareness, technical value, learning value, etc.

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Create a the Product Backlog



- Given a variety of “desirements” identify and order the PBIs into a meaningful product backlog
 1. Each table must select a Product Owner.
 2. Go through the provided list and determine which are Product Backlog items, acceptance criteria, or noise.
 3. Record the PBIs (and any acceptance criteria) on sticky notes.
 4. Product Owner should provide a business value and order the PBIs.
 5. Be prepared to defend your decisions.

15 Minutes

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Product Backlog Grooming

- Grooming is the act of adding detail, estimates, and order to items in the Product Backlog
 - This is an ongoing process in which the Product Owner and the Development Team collaborate on the details of Product Backlog items
- The Scrum Guide says grooming can (should) take up to 10% of the length of Sprint
 - Budget the time, schedule the meeting, and attend

Agile Estimation

- To be more accurate, be less precise
 - Estimate smaller things (tasks) in hours
 - Estimate larger things (PBIs/Bugs) in something less precise/abstract
- You should use an abstract unit of measure
 - Fibonacci, story points, acorns, kazoobies, Vicodin
 - T-shirt sizes (S, M, L, XL)
- Planning Poker®

Agile Estimation

- The whole development team participates
 - The Product Owner and Scrum Master do not (unless they are also Development Team members)
- Estimate as late as is responsible
 - Scrum offers two formal opportunities for estimation: Sprint Planning and Product Backlog grooming
- Estimating undesirable PBIs is waste
 - Wait until larger (Epic) Product Backlog items are decomposed
 - Wait until the Product Owner orders (prioritizes) the Product Backlog

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Estimate Using Planning Poker



- Assuming Spain is a 3, estimate the land area of the following countries relative to Spain:
 1. Luxembourg
 2. Denmark
 3. Belize
 4. South Africa
 5. China
 6. United States

Note: Use 1, 2, 3, 5, 8, 13, 21, 34, 55

5 Minutes

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Estimation is Not a Silver Bullet

- Agile estimation techniques won't remove uncertainty from early estimates
 - They will improve your accuracy as the project proceeds
 - This is because Agile estimation methods take actual work into account as Sprints are completed

Begin Sprinting

- Just get started
 - The team can inspect & adapt its progress during the Sprint
- Be able to assess progress
- Inspect and Adapt

How long should the Sprint be?

"As short as possible, and no shorter"



Use a Task Board to Assess Progress

PBI	To-Do	In-Progress	Done
			   
	 	 	 
	  		
	    		

"Done"

- Done defines when a Product Backlog item is 100% complete and potentially releasable
- Definition of "Done"
 - An auditable checklist the Development Team uses to know when they are done developing each PBI
 - The definition should be clearly understood by the Product Owner

Example Definition of "Done"

- ✓ All tests pass
- ✓ All code compiles
- ✓ All builds automated
- ✓ All code written using TDD
- ✓ An .msi installer is created

Continuous Improvement

The Culture Must Change

- An organization's culture is finely tuned to produce its current ~~conditions~~ problems
- Agility is an entirely new state
 - Culture must change to achieve Agility
- "I want to run an agile project"
 - <http://bit.ly/jBB5RO>



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Overcome Dysfunction

Dysfunctional behaviors can be found everywhere:

- | | |
|------------------------------|----------------------------|
| • Scrum is a "Silver Bullet" | • Challenging Stakeholders |
| • Scrum Master Issues | • Definition of Done |
| • Product Owner Issues | • Sprint Backlog Issues |
| • Team Issues | • Velocity Issues |
| • Working as a Team | • Retrospective Issues |

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What about Dieter?



As a team, how would you handle Dieter?



Dieter is your organization's only SQL Server administrator. He holds all the passwords and all database changes must go through him. He's happy to help your team with its database tasks, but only for a few days. He's very busy, and important. He makes it clear that he doesn't work for you or your boss and his help is just a courtesy.

5 Minutes

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What about Roger?



As a team, how would you handle Roger?



Roger is your team's Scrum Master, but his real background is in project management. He is getting pressure from his managers to provide more accurate costs for the software development project. To that end, he is asking that you start tracking the actual hours you spend on tasks.

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Get a Good Scrum Master

- The Scrum Master
 - Knows Scrum and ensures rules are followed
 - Shields team from distractions
 - Helps remove impediments
 - Ensures the team is fully functional, productive, and improves quality
 - Enables cooperation across all roles/functions
 - Removes barriers
 - Educates everyone on Scrum
 - Is a de facto coach

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Remember: Scrum Doesn't Fail



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- Scrum is like chess
 - You either play it as its rules state, or you don't
- Scrum and chess do not fail or succeed
 - They are either played, or not
- Read the "*Scrum Fails?*" blog post
 - <http://bit.ly/fOcrct>

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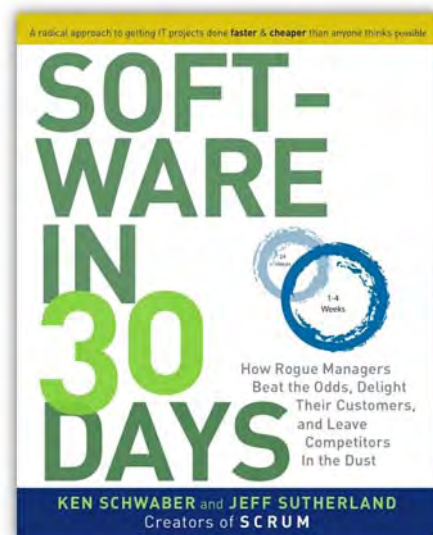
Next Steps

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Read Ken's Book

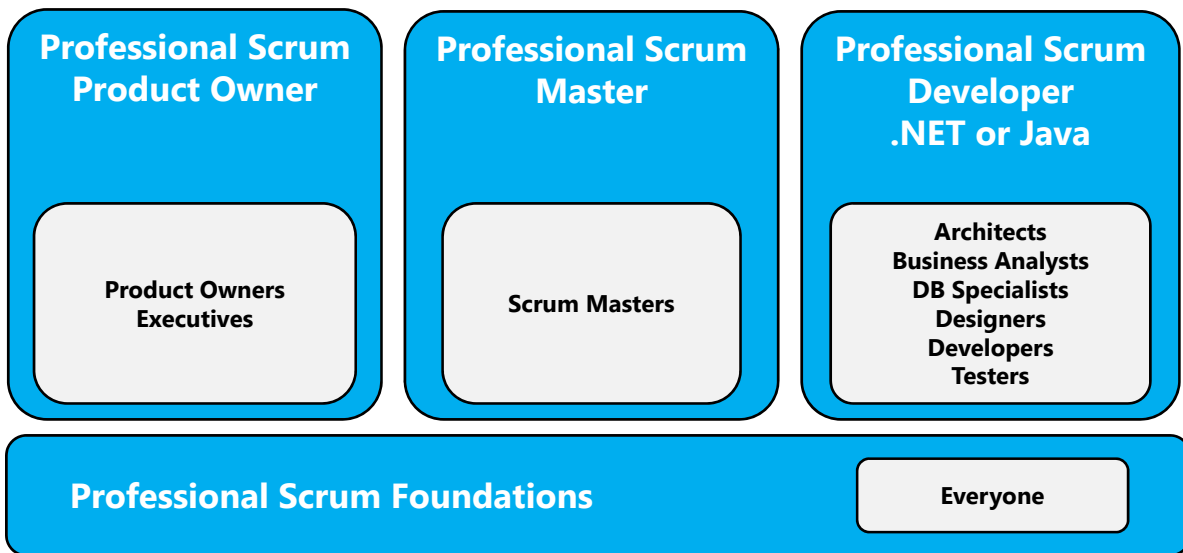
- Software in 30 Days
 - By Ken Schwaber and Jeff Sutherland
 - Available now



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Be Professional



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Be Part of the Community



Follow @scrumdotorg and search for the #scrum and #proscrumdev hashtag on Twitter



Join the Yahoo! discussion group where you can tap into the collective knowledge of Scrum users around the world:
<http://groups.yahoo.com/group/scrumdevelopment>



Join the AgileBoise group on LinkedIn:
<http://www.linkedin.com/groups/AgileBoise-1683377>

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Done () ;

(thank you)

